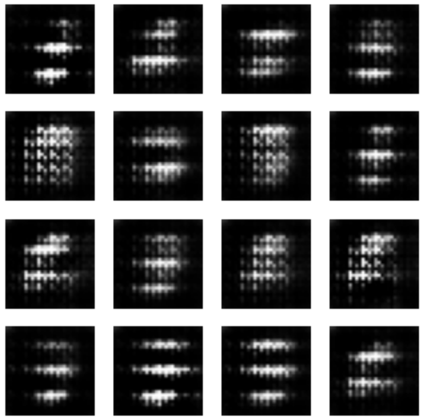

EXPRIEMENT 4A

```
# ==============================
# 1️⃣ Install Required Libraries
# ==============================
!pip install --quiet transformers torch

# ==============================
# 2️⃣ Import Libraries
# ==============================
from transformers import pipeline
from IPython.display import display, Markdown

# ==============================
# 3️⃣ Load Models (All Local)
# ==============================
gpt2 = pipeline("text-generation", model="gpt2")
distilgpt2 = pipeline("text-generation", model="distilgpt2")
gpt2_medium = pipeline("text-generation", model="gpt2-medium")

# ==============================
# 4️⃣ Common Prompt
# ==============================
prompt = "Write a haiku about AI and nature."

# ==============================
# 5️⃣ Generate Responses
# ==============================
competitors = [
    "GPT2",
    "DistilGPT2",
    "GPT2-Medium"
]

answers = [
    gpt2(prompt, max_length=60, temperature=0.7)[0]["generated_text"],
    distilgpt2(prompt, max_length=60, temperature=0.7)[0]["generated_text"],
    gpt2_medium(prompt, max_length=60, temperature=0.7)[0]["generated_text"]
]

# ==============================
# 6️⃣ Display Outputs
# ==============================
display(Markdown("## 📌 Model Outputs\n"))

for comp, ans in zip(competitors, answers):
    display(Markdown(f"### 🤖 {comp}\n{ans}\n---"))

# ==============================
# 7️⃣ Simple Automatic Evaluation
# ==============================
def score_response(text):
    score = 0
    score += len(text.split())  # length
    score += text.count("\n") * 5  # poetic structure bonus
    return score
```

```
scores = [(comp, score_response(ans)) for comp, ans in zip(competitors, answers)]
scores.sort(key=lambda x: x[1], reverse=True)

# ==============================
# 8️⃣ Display Ranking
# ==============================
display(Markdown("## 🏆 Final Ranking (Best → Worst)\n"))

for i, (model, score) in enumerate(scores):
    display(Markdown(f"**Rank {i+1}: {model}** — Score: {score}"))
```

Loading weights: 100% 148/148 [00:00<00:00, 709.33it/s, Materializing param=transformer.wte.weight]

```
GPT2LMHeadModel LOAD REPORT from: gpt2
Key                  | Status     |  |
---------------------+------------+--+-
h.{0...11}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

Loading weights: 100% 76/76 [00:00<00:00, 411.40it/s, Materializing param=transformer.wte.weight]

```
GPT2LMHeadModel LOAD REPORT from: distilgpt2
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
transformer.h.{0, 1, 2, 3, 4, 5}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

Loading weights: 100% 292/292 [00:00<00:00, 698.30it/s, Materializing param=transformer.wte.weight]

```
GPT2LMHeadModel LOAD REPORT from: gpt2-medium
Key                  | Status     |  |
---------------------+------------+--+-
h.{0...23}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
```

Start coding or generate with AI.

```
Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer t
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer t
```

📌 Model Outputs

🤖 GPT2

Write a haiku about AI and nature.

"Humans have evolved since ancient times to become the most intelligent beings on the planet," said Dr. David N. Friedman of the Stanford University School of Medicine in a press release. "In the past, this has been an issue for us. Humans have been working on this problem for many decades, but now that we have made progress it's time to take a step back and realize that we are going to change things. We are not going to be able to solve this problem from a technological standpoint. So we need to find ways to communicate with each other and to solve the problem, and I am very optimistic about this project."

This is the fifth time the Stanford team has used a "mind-controlled" robot, and it was created by the University of Texas at Austin, Dr. David Friedman said.

"It's probably the most powerful robot in the world," he said. "It can act like a human and use its own emotions to communicate with other robots. It can even act like a robot, so it can communicate with itself and it can even communicate with other robots. It's a really beautiful